# JDBC & Hibernate Interview Questions and Answers

## JDBC Questions

### *What is JDBC?*

JDBC is a Java API used to connect Java applications with databases. It provides classes and interfaces for executing SQL queries. It acts as a bridge between Java and DBMS.

### *What are the JDBC components?*

Driver, Connection, Statement, ResultSet, and DriverManager are core JDBC components. They help establish connections and execute SQL. Together they enable database operations.

### *What is DriverManager?*

DriverManager manages JDBC drivers and creates database connections. It selects the appropriate driver automatically. It is commonly used with getConnection().

### *What is a JDBC Driver?*

A JDBC Driver is software that enables Java applications to communicate with databases. It converts JDBC calls into DB-specific calls. Different databases provide different drivers.

### *Steps to connect database using JDBC?*

Load driver, establish connection, create statement, execute query, and close resources. This is the standard JDBC workflow. Proper exception handling is recommended.

### *Statement vs PreparedStatement?*

Statement executes static SQL queries. PreparedStatement supports parameterized queries and prevents SQL injection. PreparedStatement is faster for repeated execution.

### *What is ResultSet?*

ResultSet stores data returned by a SELECT query. It allows traversal through rows and columns. Methods like next() are used to iterate.

### *What is SQL Injection?*

SQL Injection is a security vulnerability caused by concatenating user input into SQL. Attackers can manipulate queries. PreparedStatement helps prevent it.

### *What is Batch Processing?*

Batch processing executes multiple SQL statements together. It improves performance by reducing database round trips. It is useful for bulk inserts and updates.

### *What is Connection Pooling?*

Connection pooling reuses existing database connections. It improves application performance and scalability. Popular implementations include HikariCP and C3P0.

# Hibernate Questions

### *What is Hibernate?*

Hibernate is an ORM framework for Java. It maps Java objects to database tables. It reduces the need for writing SQL manually.

### *What is ORM?*

ORM stands for Object Relational Mapping. It maps Java classes to database tables. This simplifies database interactions.

### *Advantages of Hibernate?*

Hibernate reduces boilerplate JDBC code. It provides caching, lazy loading, and database independence. Development becomes faster and easier.

### *What is Session in Hibernate?*

Session is the primary interface for interacting with Hibernate. It represents a single unit of work. CRUD operations are performed through it.

### *Session vs SessionFactory?*

SessionFactory is heavyweight and created once per application. Session is lightweight and created per request. SessionFactory creates Sessions.

### *What is HQL?*

HQL stands for Hibernate Query Language. It is object-oriented and works with entities instead of tables. It is database independent.

### *What are Hibernate Annotations?*

Annotations provide metadata for mapping classes to tables. Examples include @Entity, @Table, and @Id. They replace XML configurations.

### *What is Lazy Loading?*

Lazy loading delays object loading until needed. It improves performance by fetching data on demand. Hibernate supports it by default for associations.

### *What is Eager Loading?*

Eager loading fetches associated data immediately. It reduces later database calls. However, it may affect performance if overused.

### *First Level Cache?*

First-level cache is enabled by default in Hibernate. It exists at Session level. It reduces repeated database access within the same session.

### *Second Level Cache?*

Second-level cache exists at SessionFactory level. It is shared across sessions. It improves application performance significantly.

### *get() vs load()?*

get() immediately fetches data and returns null if not found. load() returns a proxy object. load() throws an exception if data is missing.

### *Transient, Persistent, Detached States?*

Transient objects are not associated with Hibernate. Persistent objects are managed by Session. Detached objects were persistent but session is closed.

### *What is Cascade?*

Cascade propagates operations from parent to child entities. Examples include save, update, and delete. It simplifies entity management.

### *What is N+1 Problem?*

N+1 occurs when one query loads parent records and multiple queries load child records. It causes performance issues. Fetch joins can solve it.